JOURNAL OF THE AMERICAN COLLEGE OF CARDIOLOGY VOL. 84, NO. 3, 2024
© 2024 THE AUTHORS. PUBLISHED BY ELSEVIER ON BEHALF OF THE AMERICAN COLLEGE OF CARDIOLOGY FOUNDATION. THIS IS AN OPEN ACCESS ARTICLE UNDER THE CC BY-NC-ND LICENSE (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Semaglutide and NYHA Functional Class in Obesity-Related Heart Failure With Preserved Ejection Fraction

The STEP-HFpEF Program

Morten Schou, MD,[a,*] Mark C. Petrie, MD,[b,*] Barry A. Borlaug, MD,[c] Javed Butler, MD,[d,e] Melanie J. Davies, MD,[f,g] Dalane W. Kitzman, MD,[h] Sanjiv J. Shah, MD,[i] Subodh Verma, MD, PhD,[j] Shachi Patel, PhD,[k] Khaja M. Chinnakondepalli, MS,[k] Signe Harring, MD PhD,[l] Steen Z. Abildstrøm, PhD,[l] Karoline Liisberg, PhD,[l] Mikhail N. Kosiborod, MD,[m] the STEP-HFpEF Trial Committees and Investigators

ABSTRACT

BACKGROUND In the Semaglutide Treatment Effect in People with obesity and HFpEF (STEP-HFpEF) program, semaglutide improved heart failure (HF)-related symptoms, physical limitations, and exercise function, and reduced bodyweight in patients with obesity-related heart failure with preserved ejection fraction (HFpEF). Whether semaglutide improves functional status, as assessed by NYHA functional class, is unknown.

OBJECTIVES The goal of this study was to examine the effects of semaglutide on change in NYHA functional class over time. We also investigated the effects of semaglutide on HF-related symptoms, physical limitations, and bodyweight and other trial endpoints across baseline NYHA functional class categories.

METHODS This was a prespecified analysis of pooled data from 2 international, double-blind, randomized trials (STEP-HFpEF and STEP-HFpEF type 2 diabetes [STEP-HFpEF DM], comprising the STEP-HFpEF program), which collectively randomized 1,145 participants with obesity-related HFpEF to once-weekly semaglutide 2.4 mg or placebo for 52 weeks. The outcome of interest for this analysis was the change in NYHA functional class (baseline to 52 weeks). We also investigated the effects of semaglutide on the dual primary, confirmatory secondary, and selected exploratory endpoints according to baseline NYHA functional class.

RESULTS More semaglutide-treated than placebo-treated patients had an improvement in NYHA functional class (32.6% vs 21.5%, respectively; OR: 2.20 [95% CI: 1.62-2.99; $P < 0.001$]) and fewer semaglutide-treated patients experienced deterioration in NYHA functional class (2.09% vs 5.24%, respectively; OR: 0.36 [95% CI: 0.19-0.70; $P = 0.003$]) at 52 weeks. Semaglutide (vs placebo) improved the Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score (KCCQ-CCS) across NYHA functional class categories; this was especially pronounced in those in NYHA functional classes III/IV (10.5 points [95% CI: 6.6-14.4 points]) vs NYHA functional class II (6.0 points [95% CI: 3.4-8.6 points]) (P interaction = 0.06). By contrast, the degree of reduction in bodyweight was similar with semaglutide vs placebo regardless of baseline NYHA functional class category (NYHA functional class II, −8.4% [95% CI: −9.4% to −7.3%]; NYHA functional classes III/IV, −8.3% [95% CI: −9.9% to −6.8%]; P interaction = 0.96). Semaglutide consistently improved 6-minute walking distance (6MWD), the hierarchical composite endpoint (death, HF events, differences in KCCQ-CSS, and 6MWD changes), and reduced C-reactive protein and N-terminal prohormone of brain natriuretic peptide across NYHA functional class categories (all P interactions = NS).

CONCLUSIONS In patients with obesity-related HFpEF, fewer semaglutide-treated than placebo-treated patients had a deterioration, and more had an improvement, in NYHA functional class at 52 weeks. Semaglutide consistently improved HF-related symptoms, physical limitations, and exercise function, and reduced bodyweight and biomarkers of inflammation and congestion in all NYHA functional class categories. Semaglutide-mediated improvements in health status were especially large in patients with NYHA functional classes III/IV. (Research Study to Look at How Well Semaglutide Works in People Living With Heart Failure and Obesity; NCT04788511) (Research Study to Look at How Well Semaglutide Works in People Living With Heart Failure, Obesity and Type 2 Diabetes; NCT04916470) (J Am Coll Cardiol 2024;84:247-257) © 2024 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Listen to this manuscript's audio summary by Editor Emeritus Dr Valentin Fuster on www.jacc.org/journal/jacc.

ISSN 0735-1097 https://doi.org/10.1016/j.jacc.2024.04.038

JACC VOL. 84, NO. 3, 2024
JULY 16, 2024:247–257

ABBREVIATIONS AND ACRONYMS

6MWD = 6-minute walking distance

BMI = body mass index

CRP = C-reactive protein

GLP-1RA = glucagon-like peptide-1 receptor agonists

HF = heart failure

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

hsCRP = high-sensitivity C-reactive protein

KCCQ-CSS = Kansas City Cardiomyopathy Questionnaire Clinical Summary Score

NT-proBNP = N-terminal prohormone of brain natriuretic peptide

SAEs = serious adverse events

SGLT2 = sodium-glucose cotransporter 2

One of the main treatment goals in patients with heart failure (HF) with preserved ejection fraction (HFpEF) is to improve functional status. The need for novel therapies that can accomplish this goal is most pressing in patients with obesity-related HFpEF, who have a particularly profound impairment in their functional status.[1] Currently, sodium-glucose cotransporter 2 (SGLT2) inhibitors represent the only guideline-recommended class of agents shown to improve functional status in patients with HFpEF.[2,3] However, the functional benefits of SGLT2 inhibitors in this patient group are relatively modest, with a persistent clinical need for additional efficacious therapies.

SEE PAGE 258

In the Semaglutide Treatment Effect in People with obesity and HFpEF (STEP-HFpEF) program,[4-6] once-weekly semaglutide produced large improvements in HF-related patient-reported symptoms and physical limitations (assessed by the Kansas City Cardiomyopathy Questionnaire Clinical Summary Score [KCCQ-CSS]), improved exercise function, and reduced bodyweight in patients with obesity-related HFpEF (both with and without type 2 diabetes).[6] In clinical practice, functional status is conventionally assessed by clinicians using the NYHA functional class. Whether treatment with semaglutide is effective in improving clinician-assessed NYHA functional classification over time is unknown.

Evaluating the effects of therapies across different NYHA functional class categories, as well as their effects on NYHA functional class in patients with HFpEF, is important for several reasons. First, despite acknowledged limitations,[7-10] the NYHA functional class is a simple, quick, and easy prognostic tool, which remains widely used in clinical practice (and in HF trials) both for assessment of functional status and for risk stratification. Importantly, patients with HF in higher NYHA functional classes are not only at higher risk of clinical deterioration and adverse events, but also may be less likely to respond to some HF therapies. For example, sacubitril/valsartan may not be as effective in patients with HF with reduced ejection fraction (HFrEF) in NYHA functional classes III/IV as it is in NYHA functional class II.[11] NYHA functional class assessment is central to many recommendations in international HF practice guidelines.[2,3] Furthermore, although NYHA functional class assessments may be less sensitive to clinical change than patient-reported outcomes (such as KCCQ), demonstration of therapeutic benefit on clinician-assessed NYHA functional class, in addition to KCCQ, would further buttress the potential value of a therapeutic intervention.

In this prespecified analysis of the STEP-HFpEF program, our main objective was to examine the effects of semaglutide on changes in NYHA functional class over time. We also investigated the effects of semaglutide on the dual primary (change in KCCQ-CSS and bodyweight at 52 weeks) and confirmatory secondary endpoints as well as an exploratory endpoint, N-terminal prohormone of brain natriuretic peptide (NT-proBNP), of the STEP-HFpEF program according to NYHA functional class at baseline.

METHODS

PROGRAM DESIGN. This is a prespecified analysis of the international, multicenter, double-blind, randomized, placebo-controlled STEP-HFpEF program (comprising the STEP-HFpEF and STEP-HFpEF DM

From the [a]Department of Cardiology, Herlev-Gentofte Hospital, University of Copenhagen, Herlev, Denmark; [b]School of Cardiovascular and Metabolic Health, University of Glasgow, Glasgow, United Kingdom; [c]Department of Cardiovascular Medicine, Mayo Clinic, Rochester, Minnesota, USA; [d]Baylor Scott and White Research Institute, Dallas, Texas, USA; [e]Department of Medicine, University of Mississippi, Jackson, Mississippi, USA; [f]Diabetes Research Centre, University of Leicester, Leicester, United Kingdom; [g]NIHR Leicester Biomedical Research Centre, Leicester, United Kingdom; [h]Section of Cardiovascular Medicine and Section on Geriatrics and Gerontology, Wake Forest University School of Medicine, Winston-Salem, North Carolina, USA; [i]Division of Cardiology, Department of Medicine, Northwestern University Feinberg School of Medicine, Chicago, Illinois, USA; [j]Division of Cardiac Surgery, Li Ka Shing Knowledge Institute of St Michael's Hospital, Unity Health Toronto, University of Toronto, Toronto, Ontario, Canada; [k]Health Economics and Technology Assessment Group, Saint Luke's Mid America Heart Institute, University of Missouri–Kansas City School of Medicine, Kansas City, Missouri, USA; [l]Novo Nordisk A/S, Søborg, Denmark; and the [m]Department of Cardiovascular Disease, Saint Luke's Mid America Heart Institute, University of Missouri–Kansas City School of Medicine, Kansas City, Missouri, USA. *Drs Schou and Petrie contributed equally to this work.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received April 5, 2024; revised manuscript received April 26, 2024, accepted April 26, 2024.

[STEP-HFpEF type 2 diabetes] trials). The design and primary results of the individual trials and the overall STEP-HFpEF program have been previously published.[4-6,12] In summary, the program consisted of 2 separate trials: 1 involving patients with obesity-related HFpEF without type 2 diabetes (STEP-HFpEF; NCT04788511) and the other including patients with obesity-related HFpEF and type 2 diabetes (STEP-HFpEF DM; NCT04916470). The program was conducted at 129 sites across 18 countries in Asia, Europe, and North and South America. The steering committee, which included both academic members and representatives from the sponsor (Novo Nordisk), designed both trials and was responsible for the academic publications. A global expert panel provided academic, medical, and operational input in each country. Institutional Review Board ethics approval was obtained at each study site, and all patients provided informed consent to participate in the trial. The sponsor of the program was Novo Nordisk.

PARTICIPANTS. Participants were eligible if they had a left ventricular ejection fraction (LVEF) of ≥45%, body mass index (BMI) of ≥30 kg/m^2, NYHA functional class II/IV, KCCQ-CSS <90 points, and ≥1 of the following: 1) elevated filling pressures (based on right heart catheterization or remote pulmonary artery pressure sensor technology); 2) elevated natriuretic peptide levels (with thresholds stratified based on BMI) plus echocardiographic abnormalities; or 3) HF hospitalization in the previous 12 months plus requirement for ongoing diuretic treatment and/or echocardiographic abnormalities.

Key exclusion criteria were prior or planned bariatric surgery, self-reported change in bodyweight >11 pounds (5 kg) within 90 days before randomization, or a systolic blood pressure of >160 mm Hg at screening. In STEP-HFpEF DM, patients with uncontrolled diabetic retinopathy or maculopathy were also excluded.

TABLE 1 Baseline Characteristics According to NYHA Functional Class in the STEP-HFpEF Program

	NYHA Functional Class		
	II (n = 785)	III/IV (n = 360)	*P* Value
Female	360 (45.9)	210 (58.3)	<0.001
Age, y			<0.001
<65	272 (34.6)	96 (26.7)	
65-79	452 (57.6)	214 (59.4)	
≥80	61 (7.8)	50 (13.9)	
Ethnicity/race			<0.001
Asian	68 (8.7)	8 (2.2)	
Black or African American	33 (4.2)	6 (1.7)	
Other	2 (0.3)	2 (0.6)	
White	682 (86.9)	344 (95.6)	
Bodyweight, kg	102.1 (90.4-116.8)	106.8 (94.8-122.9)	<0.001
BMI, kg/m^2	37.2 (33.9-41.4)	40.3 (36.2-44.4)	<0.001
Waist circumference, cm	118.5 (110.0-127.0)	123.0 (114.4-134.0)	<0.001
Systolic blood pressure, mm Hg	133.0 (123.0-144.0)	134.0 (122.0-145.0)	0.864
LVEF, %	56.0 (50.0-60.0)	57.0 (50.0-61.0)	0.718
KCCQ-CSS	64.6 (50.5-76.0)	44.8 (34.3-59.4)	<0.001
6MWD, min	316.0 (241.3-382.0)	254.1 (180.0-332.1)	<0.001
hsCRP, mg/L	3.4 (1.7-7.2)	4.4 (2.0-9.8)	0.028
NT-proBNP, pg/mL	433.0 (221.6-970.0)	563.7 (288.8-1,139.6)	0.253
Comorbidities at screening			
Hypertension	653 (83.2)	306 (85.0)	0.439
Atrial fibrillation	343 (43.7)	175 (48.6)	0.120
Coronary artery disease	322 (41.0)	131 (36.4)	0.136
Concomitant medications			
Diuretics	614 (78.2)	311 (86.4)	0.001
Loop diuretics	444 (56.6)	258 (71.7)	<0.001
Thiazides	122 (15.5)	53 (14.7)	0.720
β-blockers	629 (80.1)	299 (83.1)	0.240
SGLT2 inhibitors	150 (19.1)	71 (19.7)	0.806
MRA	242 (30.8)	142 (39.4)	0.004
ACE inhibitor/ARB (ARNI)	614 (78.2)	285 (79.2)	0.716
ARNI	44 (5.6)	14 (3.9)	0.218
Insulin and analogs	81 (10.3)	47 (13.1)	0.172

Values are n (%) or median (Q1-Q3).

6MWD = 6-minute walking distance; ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; BMI = body mass index; hsCRP = high-sensitivity C-reactive protein; KCCQ-CSS = Kansas City Cardiomyopathy Questionnaire Clinical Summary Score; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal prohormone of brain natriuretic peptide; SGLT2 = sodium-glucose cotransporter 2.

RANDOMIZATION AND TRIAL PROCEDURES. Eligible participants were randomized 1:1 to receive a once-weekly target dose of semaglutide 2.4 mg subcutaneously or matching placebo on top of standard of care for 52 weeks. Randomization was stratified by BMI <35 kg/m^2 vs ≥35 kg/m^2. Among patients with type 2 diabetes enrolled in the STEP-HFpEF DM trial, semaglutide or placebo was added to background glucose-lowering medications, which could comprise any drug other than glucagon-like peptide-1 receptor agonists (GLP-1RA). Modification of glucose-lowering treatment was at the discretion of the investigator. Specific guidance regarding the adjustment of sulfonylurea and insulin doses was provided to mitigate the risk of hypoglycemia.

OUTCOMES. The dual primary endpoints of the STEP-HFpEF program were as follows: 1) change in KCCQ-CSS from baseline to 52 weeks; and 2) percent change in bodyweight from baseline to 52 weeks. The confirmatory secondary endpoints were change in 6-minute walking distance (6MWD) from baseline to 52 weeks, hierarchical composite endpoint comprising all-cause death, number and timing of adjudicated HF events (HF hospitalizations and urgent visits requiring intravenous therapy),

TABLE 2 Effect of Semaglutide vs Placebo on NYHA Functional Class

	Improvement in NYHA Functional Class		Deterioration in NYHA Functional Class	
	Week 20	Week 52	Week 20	Week 52
Semaglutide	22.51	32.64	3.32	2.09
Placebo	17.13	21.50	2.45	5.24
OR [95% CI]	1.79 [1.27-2.52]	2.20 [1.62-2.99]	1.26 [0.64-2.50]	0.36 [0.19-0.70]
P value	<0.001	<0.001	0.500	0.003

Values are % or OR [95% CI].

differences in KCCQ-CSS change (from baseline to 52 weeks) of ≥15, ≥10, and ≥5 points, and difference in 6MWD change (of ≥30 minutes from baseline to 52 weeks); and change in high-sensitivity C-reactive protein (hsCRP) from baseline to 52 weeks. HF events were adjudicated by a blinded clinical events committee as previously described.[4-6,12]

The primary aim of the current analysis was to investigate the effects of semaglutide 2.4 mg once weekly, compared with placebo, on change in NYHA functional class from baseline to 52 weeks, which was a prespecified exploratory endpoint.

Efficacy endpoints were evaluated using the full analysis set, comprising all randomized participants adhering to the intention-to-treat principle throughout the trial, irrespective of treatment discontinuation. Safety endpoints, stratified by NYHA functional class subgroups, were analyzed using the safety analysis set, encompassing all randomized participants exposed to at least 1 dose of the randomized treatment. This analysis included both on-treatment and in-trial data sets, depending on the nature of the safety event. Safety endpoints in the current analysis were serious adverse events (SAEs), SAEs leading to permanent treatment discontinuation, cardiac and gastrointestinal SAEs, and gastrointestinal adverse events leading to discontinuation of study drug.

ASSESSMENT OF NYHA FUNCTIONAL CLASS. NYHA functional class was a prespecified subgroup within the STEP-HFpEF program, with the 2 selected categories being NYHA functional class II and NYHA functional classes III and IV (ie, III and IV combined in 1 group). Study investigators, who were blinded to treatment allocation, assessed and reported NYHA functional class at baseline, and 20 and 52 weeks.

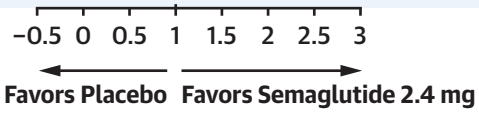
STATISTICAL ANALYSES. Baseline characteristics were assessed based on NYHA functional class (II vs III/IV). Student's *t*-test was employed to compare the continuous variables, while chi-square or Fisher exact test were used for the categorical variables. Efficacy endpoints were evaluated using the full analysis set, comprising all randomized participants adhering to the intention-to-treat principle throughout the trial, irrespective of treatment discontinuation. Safety endpoints, stratified by NYHA functional class subgroups, were analyzed using the safety analysis set, encompassing all randomized participants exposed to at least 1 dose of the randomized treatment. This analysis included both on-treatment and in-trial data sets, depending on the nature of the safety event.

For the change in KCCQ-CSS and 6MWD, missing observations at week 52 due to cardiovascular death or previous HF events were imputed singularly to the lowest observed value across both treatment arms and visits. Missing values for other reasons were multiple imputed using data from participants in the same randomized treatment arm. In cases of other endpoints, missing observations at week 52 were multiple imputed, regardless of death or prior HF events, utilizing the same imputation method. Effects of semaglutide vs placebo on the continuous endpoints across the NYHA functional class subgroups were performed using analysis of covariance models, with the change in the corresponding endpoint at week 52 as the dependent variable. Fixed factors included randomized treatment, trial, BMI (<35 kg/m^2 or ≥35 kg/m^2), NYHA functional class subgroup, and treatment by subgroup interaction, adjusted for the baseline value of the corresponding endpoint across 1,000 imputation data sets. Estimates were combined using Rubin's rule, and interaction *P* values were derived from an F-test assessing equality between the treatment differences across the NYHA functional class subgroups.

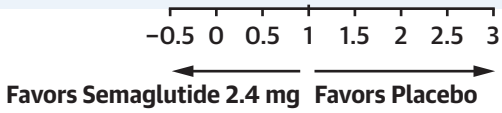
Analyses of the hierarchical composite endpoint (win ratio) compared each participant randomized to semaglutide against each participant randomized to placebo. For each of the participant pairs, a "treatment winner" based on similar observation times was declared based on the endpoint hierarchy. The win ratio (ie, the proportion of winners randomized to semaglutide divided by the winners randomized to placebo) was estimated using 1,000 imputations as described in the previous text. The win ratio was estimated independently within each NYHA functional classification subgroup. Cochran's Q test was used to test for equality of the NYHA functional classification subgroups for the win ratio.

FIGURE 1 Effects of Semaglutide vs Placebo on NYHA Functional Class

NYHA Functional Class Improvement	Semaglutide 2.4 mg, n (%)	Placebo, n (%)	OR (95% CI)	*P* Value
Week 20	129 (23.6)	98 (18.1)	1.79 (1.27-2.52)	<0.001
Week 52	187 (34.6)	123 (23.3)	2.20 (1.62-2.99)	<0.001



NYHA Functional Class Deterioration	Semaglutide 2.4 mg, n (%)	Placebo, n (%)	OR (95% CI)	*P* Value
Week 20	19 (3.5)	14 (2.6)	1.26 (0.64-2.50)	0.5
Week 52	12 (2.2)	30 (5.7)	0.36 (0.19-0.70)	0.003



P values are for the ORs for a change in NYHA functional class over time.

Treatment effects of semaglutide on NYHA functional class at each follow-up time point were assessed using the observed data from baseline, 20-week, and 52-week visits (without imputation for missing data). We examined the proportions of semaglutide-treated and placebo-treated patients experiencing at least 1 class deterioration, no change, and at least 1 class improvement in NYHA functional class between baseline and week 20, as well as baseline and week 52. Logistic regression models were then used to calculate the ORs and corresponding 95% CIs for semaglutide effects on the likelihood of ≥1 NYHA functional class deterioration, as well as ≥1 NYHA functional class improvement, separately at week 20 and week 52, adjusted for the baseline NYHA functional class, treatment and trial, and BMI group (the stratification factor).

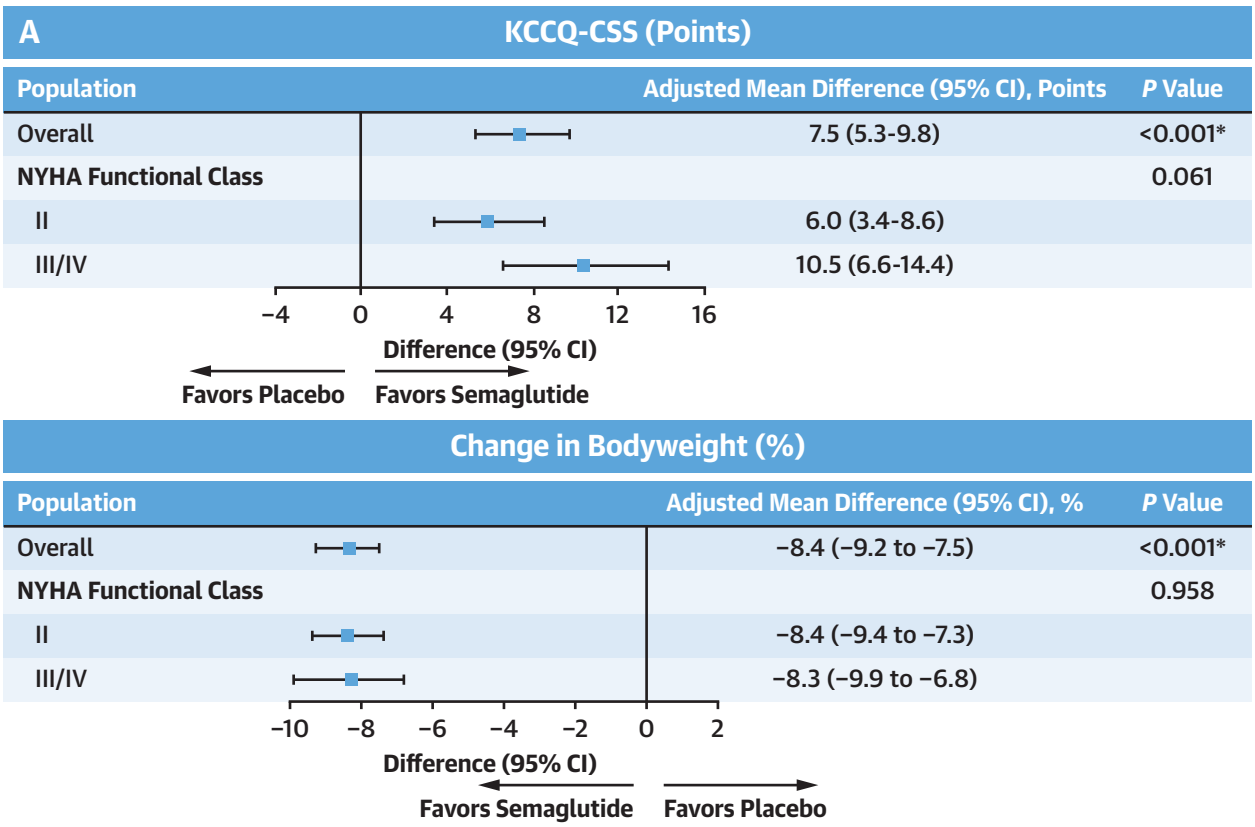
No adjustment for multiple testing was carried out, and a significance level of 5% was deemed significant. Results are presented as estimated changes from baseline to week 52 for continuous endpoints or a win ratio (for the hierarchical composite endpoint), along with a 95% CI and a 2-sided *P* value. NT-proBNP and hsCRP were log-transformed, and treatment ratios at week 52 are reported. All statistical analyses were performed using SAS version 9.4 or SAS/STAT version 15.1. All data provided were anonymized. Analyses were conducted by the independent statistical group at Saint Luke's Mid America Heart Institute in collaboration with Novo Nordisk.

RESULTS

PATIENT CHARACTERISTICS BY NYHA FUNCTIONAL CLASS. A total of 1,145 patients were randomized across 129 sites and 18 countries in the STEP-HFpEF program (Table 1). Of those, 785 patients (69%) were classified in NYHA functional class II and 360 (31%) were classified in NYHA functional classes III/IV. Of those in NYHA functional classes III/IV, 358 participants were in functional class III and 2 were in functional class IV. Those in NYHA functional classes III/IV, compared with NYHA functional class II, were more likely to be female, older, and White. Patients in NYHA functional classes III/IV had a lower baseline KCCQ-CSS and shorter 6MWD at baseline. Those in NYHA functional classes III/IV, compared with NYHA functional class II, had a higher BMI and higher CRP levels and were more frequently treated with diuretics.

EFFECT OF SEMAGLUTIDE VS PLACEBO ON NYHA FUNCTIONAL CLASS AT 20 AND 52 WEEKS. A total of 77 patients (6.7%) had a missing NYHA functional

FIGURE 2 Effects of Semaglutide vs Placebo on the STEP-HFpEF Program Endpoints Across NYHA Functional Class Categories



(A) Dual primary outcomes. (B) Secondary outcomes. (C) Exploratory outcomes. *P* values are for the test for interaction between treatment and NYHA functional class subgroups and * for the treatment difference between semaglutide and placebo. 6MWD = 6-minute walking distance; BW = bodyweight; CRP = C-reactive protein; KCCQ-CSS = Kansas City Cardiomyopathy Questionnaire Clinical Summary Score; NT-proBNP = N-terminal prohormone of brain natriuretic peptide.

Continued on the next page

class registration value at week 52 (32 in the semaglutide group, 45 in the placebo group). More semaglutide-treated patients had an improvement of at least 1 NYHA functional class or more (32.6% vs 21.5%), from baseline to 52 weeks. Fewer semaglutide-treated than placebo-treated patients experienced a deterioration of at least 1 NYHA functional class or more from baseline to 52 weeks (2.09% vs 5.24%) (Table 2). Improvement in NYHA functional class with semaglutide vs placebo was evident at 20 weeks and was further amplified at 52 weeks (Table 2).

Similarly, in the logistic regression models, semaglutide-treated vs placebo-treated patients had significantly higher odds of improvement (2.20 [95% CI: 1.62-2.99; $P < 0.001$]) and lower odds of deterioration (0.36 [95% CI: 0.19-0.70; $P = 0.003$]), of at least 1 NYHA functional class from baseline to 52 weeks. The odds of improvement in NYHA functional class with semaglutide vs placebo were seen at 20 weeks and were further amplified at 52 weeks (Figure 1).

EFFECT OF SEMAGLUTIDE VS PLACEBO ON THE PRIMARY, SECONDARY, AND EXPLORATORY ENDPOINTS BY BASELINE NYHA FUNCTIONAL CLASS. Semaglutide, compared with placebo, led to significant improvement in KCCQ-CCS across all baseline NYHA functional class categories; but this was especially pronounced among participants in NYHA functional classes III/IV (10.5 points [95% CI: 6.6-14.4 points]) than in NYHA functional class II (6.0 points [95% CI: 3.4-8.6 points]) (*P* interaction = 0.06) (Figure 2A). Semaglutide, compared with placebo, reduced bodyweight to a similar extent regardless of baseline NYHA category (NYHA functional class II −8.4% [95% CI: −9.4% to −7.3%] and NYHA functional classes III/IV −8.3% [95% CI −9.9% to −6.8%]; *P* interaction = 0.96).

FIGURE 2 Continued

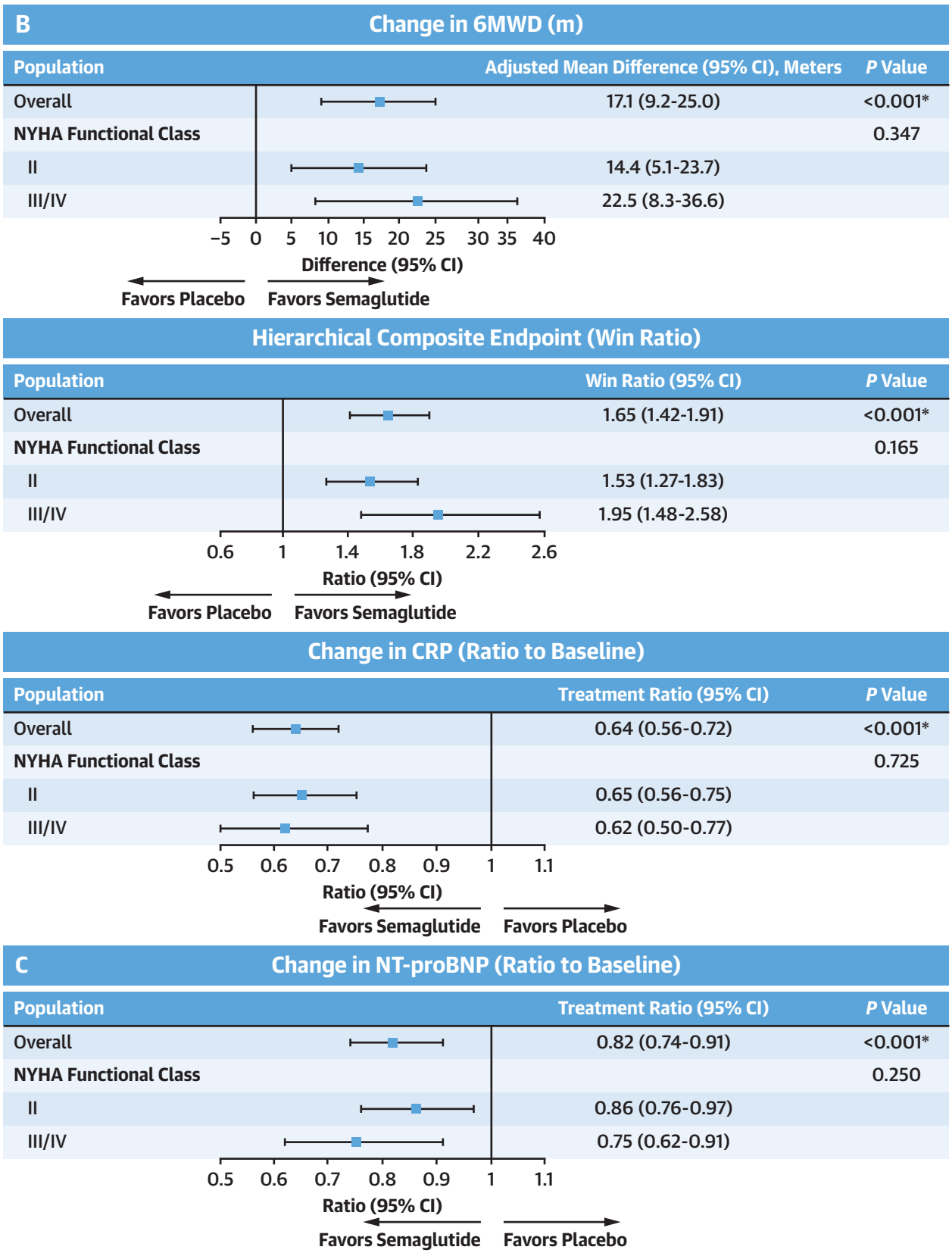


TABLE 3 Effect of Semaglutide vs Placebo on Primary and Secondary Endpoints According to Baseline NYHA Functional Class

Endpoint	NYHA Functional Class II (N = 785)		NYHA Functional Classes III/IV (N = 360)		*P* Value for Interaction
	Semaglutide (n = 406)	Placebo (n = 379)	Semaglutide (n = 167)	Placebo (n = 193)	
Dual primary endpoints					
Change in KCCQ-CSS at 52 weeks, points	15.8	9.8	13.3	2.8	
Adjusted mean difference, points	6.0 (3.4-8.6)		10.5 (6.6-14.4)		0.06
Change in bodyweight at 52 weeks, %	−11.3	−2.9	−11.6	−3.2	
Adjusted mean difference, %-points	−8.4 (−9.4 to −7.3)		−8.3 (−9.9 to −6.8)		0.96
Confirmatory secondary endpoints					
Change in 6MWD at 52 weeks, m	16.5	2.1	17.4	−5.0	
Adjusted mean difference, m	14.4 (5.1-23.7)		22.5 (8.3-36.6)		0.35
hsCRP ratio at 52 weeks	0.57	0.88	0.58	0.94	
Treatment ratio	0.65 (0.56-0.75)		0.62 (0.50-0.77)		0.76
Hierarchical endpoint					
Win ratio	1.53 (1.27-1.83)		1.95 (1.48-2.58)		0.17
NT-proBNP ratio at 52 weeks	0.76	0.89	0.82	1.09	
Treatment ratio	0.86 (0.76-0.97)		0.75 (0.62-0.91)		0.25

Abbreviations as in Table 1.

Improvement in 6MWD was consistent across baseline NYHA category (NYHA functional class II 14.4 minutes [95% CI: 5.1-23.7 minutes], NYHA functional classes III/IV 22.5 minutes [95% CI: 8.3-36.6 minutes]; *P* interaction = 0.35) (Figure 2B). Semaglutide had a similar effect on the hierarchal composite endpoint (*P* interaction = 0.17) as well as the reductions in hsCRP (*P* interaction = 0.73) and NT-proBNP (*P* interaction = 0.25) across baseline NYHA functional class categories (Table 3, Figures 2B and 2C).

SAFETY DATA BY BASELINE NYHA FUNCTIONAL CLASS. The favorable safety profile of semaglutide, compared with placebo, was consistent across NYHA functional class categories (Table 4). SAEs and serious cardiac disorders were less common with semaglutide than with placebo across NYHA functional class categories. Gastrointestinal SAEs and SAEs leading to drug discontinuation were similar in semaglutide-treated and placebo-treated patients, regardless of NYHA functional class.

TABLE 4 Adverse Events in Semaglutide and Placebo Groups Across NYHA Functional Class

Adverse Event	NYHA Functional Class II (N = 785)		NYHA Functional Classes III/IV (N = 360)	
	Semaglutide (n = 406)	Placebo (n = 379)	Semaglutide (n = 167)	Placebo (n = 193)
Serious adverse events	59 (14.5)	92 (24.3)	31 (18.6)	67 (34.7)
Cardiac serious adverse events	15 (3.7)	38 (10.0)	11 (6.6)	32 (16.6)
Gastrointestinal serious adverse events	10 (2.5)	4 (1.1)	2 (1.2)	8 (4.1)
Serious adverse events leading to premature treatment discontinuation	9 (2.2)	12 (3.2)	3 (1.8)	5 (2.6)
COVID-19-related adverse events	51 (12.6)	43 (11.4)	22 (13.2)	37 (19.2)

Values are n (%).

DISCUSSION

In this analysis of 1,145 patients with obesity-related HFpEF in the STEP-HFpEF program, once-weekly semaglutide resulted in an early (at 20 weeks) and sustained (at 52 weeks) improvement in HF functional status as assessed by NYHA functional class (Figure 1). In addition, fewer semaglutide-treated than placebo-treated patients experienced a deterioration in NYHA functional class between baseline and 52 weeks. Semaglutide consistently improved HF-related symptoms, physical limitations, and exercise function, and reduced bodyweight and biomarkers of inflammation and congestion across baseline NYHA functional class subgroups. Of note, semaglutide-mediated improvements in HF-related health status were especially large in patients in NYHA functional classes III/IV.

NYHA functional class has long been a commonly used tool in clinical practice. NYHA functional class provides a simple and rapid indication of the risk of future clinical events. NYHA functional class is also a simple "workhorse" that is used to select patients for different therapies. This reflects the nearly ubiquitous use of NYHA functional class as an inclusion criterion in randomized controlled trials. International clinical practice guidelines include NYHA functional class as a central indicator as to who should receive various recommended therapies. Patients in a higher NYHA functional class represent a more severe phenotype, which may be less

responsive to treatments. For example, sacubitril/valsartan may have an attenuated effect in patients with HFrEF in those in NYHA functional classes III/IV compared with those in NYHA functional class II. In the PARADIGM-HF (Prospective comparison of ARNI with ACEI to Determine Impact on Global Mortality and morbidity in Heart Failure) trial, the treatment effect of sacubitril/valsartan compared with enalapril was more pronounced in NYHA functional class II than in NYHA functional classes III/IV.[11] In the SCD-HeFT (Sudden Cardiac Death in Heart Failure) trial, implantable cardioverter defibrillators, compared with placebo, had a major treatment effect in those in NYHA functional class II but an attenuated effect in NYHA functional class III.[13] In this context, the finding that semaglutide has a consistent effect across the endpoints in the STEP-HFpEF program is reassuring and clinically relevant. The finding that semaglutide, while leading to a quantitatively large improvement in HF-related symptoms and physical limitations in patients with NYHA functional class II, appears to have an even greater benefit in terms of improvement in KCCQ-CSS among those with more severe functional limitations (NYHA functional classes III/IV) is especially noteworthy.

The effects of semaglutide on NYHA functional class (greater likelihood of improvement and lower likelihood of deterioration) are also clinically relevant and novel findings. Although 1 of the primary endpoints of the STEP-HFpEF program was a patient-reported outcome (KCCQ-CSS, assessing HF-related symptoms and physical limitations), the beneficial impact of semaglutide on NYHA functional class is a complementary and additive finding.[14] Whereas KCCQ-CSS requires completion of a 23-item questionnaire, NYHA functional class assignment is based on a nonstandardized assessment based on simple history taking and routine clinical assessment. KCCQ-CSS, although ubiquitous in clinical trials, is still used relatively infrequently in daily clinical practice, whereas NYHA functional class is widely performed by clinicians in outpatient clinics. Given that clinicians are notoriously reluctant to change NYHA functional class assessment over time and the NYHA functional class uses only a limited number of categories, NYHA functional class is less sensitive to change than the KCCQ (scale of 0-100).[15] That semaglutide can significantly affect NYHA functional class is, therefore, as notable as the large benefits seen in KCCQ-CSS. Furthermore, the magnitude of effect of semaglutide on NYHA functional class and KCCQ-CSS compares very favorably to other HFpEF treatments (including SGLT2 inhibitors).[16-20] For example, the odds ratio for improvement in NYHA functional class with dapagliflozin was 1.32 in the DELIVER (Dapagliflozin Evaluation to Improve the LIVEs of Patients With PReserved Ejection Fraction Heart Failure) trial and 1.37 in the EMPEROR-Preserved (EMPagliflozin outcomE tRial in Patients With chrOnic heaRt Failure With Preserved Ejection Fraction) trial at week 52, respectively, compared with an odds ratio of 2.20 with semaglutide in STEP-HFpEF.[16,17]

The only previous study to assess the effect of GLP-1RAs on NYHA functional class in HF was the LIVE (Life ImproVed by Exercise) trial, which tested an entirely different hypothesis in an entirely different patient population. In 241 patients with HFrEF (LVEF <45%) with a mean BMI of 29 (ie, a mixed population of those with normal weight, overweight, and obesity), the daily GLP-1RA, liraglutide, had no effect on NYHA functional class.[21] Notably, 30% were classified in NYHA functional class I at baseline. As there is heightened interest in the role of GLP-1RAs in HF, many more data investigating the role of different agents in this class on improving functional status in different HF subpopulations will follow. The next trial that will report the effects of incretin-based therapies in obesity-related HFpEF is the SUMMIT (Study of Tirzepatide [LY3298176] in Participants With HFpEF and Obesity) trial, which is investigating the effect of tirzepatide (a GLP-1 and glucose-dependent insulinotropic polypeptide receptor agonist). Future dedicated large HF outcome trials of incretin-based agents are also needed to investigate their effects on cardiovascular death and worsening HF events, in addition to health and functional status.

The duration of the treatment effect of semaglutide on functional status in obesity-related HFpEF is of interest. In the DELIVER trial, SGLT2 inhibition with dapagliflozin improved NYHA functional class in patients with HFpEF as early as 4 weeks, with sustained benefit evident up to 32 weeks.[16] No assessment was performed beyond 32 weeks. In the EMPEROR-Preserved trial, improvements in NYHA functional class with empagliflozin were sustained for at least 2 years.[17] Future trials should evaluate the longer-term effects of incretin-based agents on the functional status in HF. In STEP-HFpEF, although the beneficial effects of semaglutide appeared to be amplified over time, patients were followed up for 52 weeks, and we cannot be certain how long these benefits may be sustained beyond 1 year.

The mechanisms of the improvement in NYHA functional class in the STEP-HFpEF program are likely multifactorial.[22] Weight loss itself can result in improved functional status, but our data suggest that

weight-independent effects of semaglutide also contribute to the observed benefits. Specifically, there appeared to be a larger improvement in HF-related symptoms and physical limitations among patients in NYHA functional classes III/IV than those in NYHA functional class II, even though the degree of weight loss was similar across NYHA subgroups. It is also possible that a similar degree of weight loss may produce larger symptomatic and functional benefits in patients with a more severe phenotype (such as a higher NYHA functional class). We also observed a reduction in NT-proBNP with semaglutide irrespective of NYHA functional class, suggesting a favorable effect on the pathobiology of obesity-related HFpEF across the spectrum of functional limitations. A reduction in CRP was also observed across baseline NYHA functional class categories, pointing toward an anti-inflammatory mechanism. Many other properties of GLP-1RAs are likely to be contributory including blood pressure-lowering and favorable systemic and coronary vascular effects.[22]

Importantly, semaglutide was well tolerated regardless of functional limitations at baseline. SAEs were less common with semaglutide than with placebo, irrespective of baseline NYHA functional class. Serious gastrointestinal adverse events were similar regardless of NYHA category.

STUDY LIMITATIONS. The STEP-HFpEF program included only a small number of patients in NYHA functional class IV, and relatively few non-White patients were enrolled. Generalizability beyond the included trial population is therefore uncertain. The duration of treatment was 52 weeks, and although there was no signal for attenuation of benefits over time, the durability of the effects on NYHA functional class beyond 1 year cannot be ascertained. NYHA functional class data were missing in a small number of participants (~7%) at 52 weeks. As with most trials, the STEP-HFpEF program was not designed to have sufficient power to address treatment effects within subgroups, and its primary objective was not focused on hospitalizations or death. Finally, we relied on NYHA functional class reported by a single-blinded investigator at each site, and assessment of NYHA functional class can be subject to interobserver variability.

CONCLUSIONS

In patients with obesity-related HFpEF, fewer semaglutide-treated than placebo-treated patients experienced a deterioration, and more experienced an improvement, in NYHA functional class at 52 weeks. Semaglutide, compared with placebo, consistently improved HF-related symptoms, physical limitations, and exercise function, and reduced bodyweight and biomarkers of inflammation and congestion in all NYHA functional class categories. Semaglutide-mediated improvements in health status were especially large in patients in NYHA functional classes III/IV.

ACKNOWLEDGMENTS Administrative support was provided by Danielle Gentry, PhD, of Apollo, OPEN Health Communications, and funded by Novo Nordisk A/S, in accordance with Good Publication Practice guidelines.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The STEP-HFpEF program was funded by Novo Nordisk. Dr Schou has received speaker fees from AstraZeneca, Boehringer Ingelheim, Novartis, and Novo Nordisk. Dr Petri has received research funding from AstraZeneca, Boehringer Ingelheim, Boston Scientific, Medtronic, Novo Nordisk, Novartis, Pharmacosmos, Roche, and SQ Innovations; and has served on committees or consulted for AbbVie, Akero, AnaCardio, Applied Therapeutics, AstraZeneca, Bayer, Biosensors, Boehringer Ingelheim, Cardiorentis, Corvia, Eli Lilly, Horizon Therapeutics, LIB Therapeutics, Moderna, New Amsterdam, Novartis, Novo Nordisk, Pharmacosmos, Siemens, SQ Innovations, Takeda, Teikoku, and Vifor. Dr Borlaug has received research support from the National Institutes of Health (NIH) and the United States Department of Defense; has received research grant funding from AstraZeneca, Axon, GlaxoSmithKline, Medtronic, Mesoblast, Novo Nordisk, Rivus, and Tenax Therapeutics; has served as a consultant for Actelion, Amgen, Aria, Axon Therapies, BD, Boehringer Ingelheim, Cytokinetics, Edwards Lifesciences, Eli Lilly, Imbria, Janssen, Merck, NGM, Novo Nordisk, NXT, and VADovations; and is named inventor (US patent no. 10,307,179) for the tools and approach for a minimally invasive pericardial modification procedure to treat heart failure. Dr Butler is a consultant to 3live, Abbott, American Regent, Amgen, Applied Therapeutics, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Cardiac Dimension, Cardior, CVRx, Cytokinetics, Edwards Lifesciences, Element Science, Eli Lilly, Impulse Dynamics, Imbria, Innolife, Inventiva, Janssen, Lexicon Pharmaceuticals, LivaNova, Medtronics, Merck, Novartis, Novo Nordisk, Occlutech, Pfizer, Pharmacosmos, PharmaIN, Roche, Sequana, SQ Innovation, and Vifor. Dr Davies has acted as consultant, advisory board member, and speaker for Boehringer Ingelheim, Eli Lilly, Novo Nordisk, and Sanofi; has served as an advisory board member and speaker for AstraZeneca; has served as an advisory board member for Medtronic, Pfizer, and ShouTi Pharma; has served as a speaker for Amgen, Novartis, and Sanofi; and has received grants as an investigator in support of investigator-initiated trials from AstraZeneca, Boehringer Ingelheim, Eli Lilly, Janssen, Novo Nordisk, and Sanofi-Aventis. Dr Kitzman was supported in part by the Kermit Glenn Phillips II Chair in Cardiovascular Medicine and NIH grants U01AG076928; R01AG078153; R01AG045551; R01AG18915; P30AG021332; U24AG059624; and U01HL160272; and has received honoraria as a consultant for AstraZeneca, Bayer, Boehringer Ingelheim, Corvia Medical, Ketyo, Novartis, Novo Nordisk, Pfizer, and Rivus; has received grant funding from AstraZeneca, Bayer, Novartis, Novo Nordisk, Pfizer, and Rivus; and has stock ownership in Gilead Sciences. Dr Shah was supported by research grants from the U.S. National Institutes of Health (NIH; U54 HL160273; R01 HL140731; and R01 HL149423); has received research funding from AstraZeneca, Corvia, and Pfizer; and has received consulting fees from Abbott, Alleviant, Amgen, Aria CV, AstraZeneca, Axon Therapies, Bayer, Boehringer Ingelheim, Boston Scientific, BridgeBio, Bristol Myers Squibb, Corvia, Cytokinetics, Edwards Lifesciences, Eli Lilly, Eidos,

Imara, Impulse Dynamics, Intellia, Ionis, Merck, NGM Biopharmaceuticals, Novartis, Novo Nordisk, Pfizer, Prothena, Regeneron, Rivus, Sardocor, Shifamed, Tenax, Tenaya, and Ultromics. Dr Verma has received speaking honoraria and/or consulting fees from Abbott, Amarin, AstraZeneca, Bayer, Boehringer Ingelheim, Canadian Medical and Surgical Knowledge Translation Research Group, Eli Lilly, HLS Therapeutics, Janssen, Merck, Novartis, Novo Nordisk, Pfizer, PhaseBio, and TIMI. Drs Harring, Abildstrøm, and Liisberg are employees and shareholders of Novo Nordisk A/S. Dr Kosiborod has served as a consultant or on an advisory board for 35Pharma, Alnylam, Amgen, Applied Therapeutics, AstraZeneca, Bayer, Boehringer Ingelheim, Cytokinetics, Dexcom, Eli Lilly, Esperion Therapeutics, Imbria, Janssen, Lexicon Pharmaceuticals, Merck (Diabetes and Cardiovascular), Novo Nordisk, Pharmacosmos, Pfizer, Sanofi, scPharmaceuticals, Structure Therapeutics, Vifor, and Youngene Therapeutics; has received research grants from AstraZeneca, Boehringer Ingelheim, and Pfizer; holds stocks in Artera Health and Saghmos Therapeutics; has received honoraria from AstraZeneca, Boehringer Ingelheim, and Novo Nordisk; and has received other research support from AstraZeneca. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Mikhail N. Kosiborod, Saint Luke's Mid America Heart Institute, Kansas City, Missouri 64111, USA. E-mail: mkosiborod@saint-lukes.org. X handle: @MkosiborodMD, @markcpetrie20.

REFERENCES

1. Nassif ME, Windsor SL, Borlaug BA, et al. The SGLT2 inhibitor dapagliflozin in heart failure with preserved ejection fraction: a multicenter randomized trial. *Nat Med*. 2021;27:1954–1960.

2. McDonagh TA, Metra M, Adamo M, et al. 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42:3599–3726.

3. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2022;79:1757–1780.

4. Kosiborod MN, Abildstrøm SZ, Borlaug BA, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2023;389:1069–1084.

5. Kosiborod MN, Petrie MC, Borlaug BA, et al. Semaglutide in patients with obesity-related heart failure and type 2 diabetes. STEP-HFpEF DM Trial Committees and Investigators. *N Engl J Med*. 2024;390:1394–1407.

6. Butler J, Shah SJ, Petrie MC, et al. Semaglutide versus placebo in people with obesity-related heart failure with preserved ejection fraction: a pooled analysis of the STEP-HFpEF and STEP-HFpEF DM randomised trials. STEP-HFpEF Trial Committees and Investigators. *Lancet*. 2024;403:1635–1648.

7. Caraballo C, Desai NR, Mulder H, et al. Clinical implications of the New York Heart Association classification. *J Am Heart Assoc*. 2019;8:e014240.

8. Raphael C, Briscoe C, Davies J, et al. Limitations of the New York Heart Association functional classification system and self-reported walking distances in chronic heart failure. *Heart*. 2007;93:476–482.

9. Tran AT, Chan PS, Jones PG, Spertus JA. Comparison of patient self-reported health status with clinician-assigned New York Heart Association classification. *JAMA Netw Open*. 2020;3:e2014319.

10. Greene SJ, Butler J, Spertus JA, et al. Comparison of New York Heart Association class and patient-reported outcomes for heart failure with reduced ejection fraction. *JAMA Cardiol*. 2021;6:522–531.

11. McMurray JJ, Packer M, Desai AS, et al. Angiotensin-neprilysin inhibition versus enalapril in heart failure. *N Engl J Med*. 2014;371:993–1004.

12. Kosiborod MN, Abildstrøm SZ, Borlaug BA, et al. Design and baseline characteristics of STEP-HFpEF program evaluating semaglutide in patients with obesity HFpEF phenotype. *J Am Coll Cardiol HF*. 2023;11:1000–1010.

13. Bardy GH, Lee KL, Mark DB, et al. Amiodarone or an implantable cardioverter-defibrillator for congestive heart failure. *N Engl J Med*. 2005;352:225–237.

14. Kosiborod MN, Verma S, Borlaug BA, et al. Effects of semaglutide on symptoms, function, and quality of life in patients with heart failure with preserved ejection fraction and obesity: a prespecified analysis of the STEP-HFpEF trial. *Circulation*. 2024;149:204–216.

15. Spertus JA, Jones PG, Sandhu AT, Arnold SV. Interpreting the Kansas City Cardiomyopathy Questionnaire in clinical trials and clinical care: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;76:2379–2390.

16. Ostrominski JW, Vaduganathan M, Claggett BL, et al. Dapagliflozin and New York Heart Association functional class in heart failure with mildly reduced or preserved ejection fraction: the DELIVER trial. *Eur J Heart Fail*. 2022;24:1892–1901.

17. Packer M, Butler J, Zannad F, et al. Effect of empagliflozin on worsening heart failure events in patients with heart failure and preserved ejection fraction: EMPEROR-Preserved trial. *Circulation*. 2021;144:1284–1294.

18. Kosiborod MN, Bhatt AS, Claggett BL, et al. Effect of dapagliflozin on health status in patients with preserved or mildly reduced ejection fraction. *J Am Coll Cardiol*. 2023;81:460–473.

19. Siddiqi TJ, Anker SD, Filippatos G, et al. Health status across major subgroups of patients with heart failure and preserved ejection fraction. *Eur J Heart Fail*. 2023;25:1623–1631.

20. Verma S, Borlaug BA, Butler J, et al. A big STEP for treatment of heart failure with preserved ejection fraction. *Cell Metab*. 2023;35:1681–1687.

21. Jorsal A, Kistorp C, Holmager P, et al. Effect of liraglutide, a glucagon-like peptide-1 analogue, on left ventricular function in stable chronic heart failure patients with and without diabetes (LIVE)-a multicentre, double-blind, randomised, placebo-controlled trial. *Eur J Heart Fail*. 2017;19:69–77.

22. Lee MMY, Petrie MC, McMurray JJV, Sattar N. How do SGLT2 (sodium-glucose cotransporter 2) inhibitors and GLP-1 (glucagon-like peptide-1) receptor agonists reduce cardiovascular outcomes?: completed and ongoing mechanistic trials. *Arterioscler Thromb Vasc Biol*. 2020;40:506–522.

KEY WORDS heart failure, left ventricular ejection fraction, New York Heart Association functional class, obesity, semaglutide